

Differential Geometry

Mathematical Tripos Part II

July 28, 2026

Note. Knowledge of what a diffeomorphism, homotopy, and isotopy are is assumed.

1 Differential Topology

1.1 Manifolds

Definition 1.1 (Manifold). We say that $X \subseteq \mathbb{R}^n$ is a **k -dimensional manifold** if each $x \in X$ has a neighborhood $V \subseteq X$ diffeomorphic to an open set of $\mathbb{R}^k$.

Definition 1.2 (Parameterisation and Chart). A diffeomorphism $\phi : U \rightarrow V$, where U is an open set of $\mathbb{R}^k$ is a **parameterisation** of the neighborhood V . The inverse diffeomorphism $\phi^{-1} : V \rightarrow U$ is called a **chart** on V .

If we have manifolds X and Z with $Z \subseteq X$ we say that Z is a **submanifold** of X . In this case, the **codimension** of Z in X is $\dim X - \dim Z$.

Definition 1.3 (Tangent Space of a Manifold). Let $X \subseteq \mathbb{R}^n$ be a manifold, $x \in X$. Let $\phi : U \rightarrow X$ be a parameterisation with $\phi(0) = x$. The **tangent space** $T_x X$ is¹ $d\phi_0(\mathbb{R}^k)$.

Let $f : X \rightarrow Y$ be a smooth map between manifolds. We say that f is a **local diffeomorphism** at x if f maps a neighbourhood of x diffeomorphically onto a neighbourhood of $f(x)$.

Theorem 1.4 (Inverse Function Theorem). *Suppose that $f : X \rightarrow Y$ is a smooth map whose derivative df_x at the point x is an isomorphism. Then f is a local diffeomorphism at x .*

1.2 Regular values and Sard's theorem

Let $f : X \rightarrow Y$ be a smooth map between manifolds. Let C be the set of all points $x \in X$ such that $df_x : T_x X \rightarrow T_{f(x)} Y$ is not surjective.

Definition 1.5. A point in C will be called a **critical point**. A point in $f(C)$ will be called a **critical value**. A point in the complement of $f(C)$ will be called a **regular value**.

Theorem 1.6 (Preimage Theorem). *Let y be a regular value of $f : X \rightarrow Y$ with $\dim X \geq \dim Y$. Then the set $f^{-1}(y)$ is a submanifold of X with $\dim f^{-1}(y) = \dim X - \dim Y$.*

¹Viewing ϕ as a function onto $\mathbb{R}^n$.

Proof. Let $x \in f^{-1}(y)$. Since y is a regular value, the derivative df_x maps $T_x X$ onto $T_y Y$. The kernel of df_x is a subspace K of $T_x X$ of dimension $p := \dim X - \dim Y$. Suppose $X \subset \mathbb{R}^N$ and let $T : \mathbb{R}^N \rightarrow \mathbb{R}^p$ be any linear map such that $\text{Ker}(T) \cap K = \{0\}$. Consider the map $F : X \rightarrow Y \times \mathbb{R}^p$ given by

$$F(z) = (f(z), T(z)).$$

The derivative of F is given by

$$dF_x(v) = (df_x(v), T(v))$$

which is clearly nonsingular by our choice of T . By the inverse function theorem, F is a local diffeomorphism at x , i.e. F maps some neighbourhood U of x diffeomorphically onto a neighbourhood V of $(y, T(x))$. Hence F maps $f^{-1}(y) \cap U$ diffeomorphically onto $(\{y\} \times \mathbb{R}^p) \cap V$ which proves that $f^{-1}(y)$ is a manifold with $\dim f^{-1}(y) = p$. $\square$

Theorem 1.7 (Stack of Records Theorem). *Let $f : X \rightarrow Y$ be a smooth map between manifolds of the same dimension with X compact. Let y be a regular value of f and write $f^{-1}(y) = \{x_1, \dots, x_k\}$. Then there exists a neighbourhood U of y in Y such that $f^{-1}(U)$ is a disjoint union $V_1 \cup \dots \cup V_k$, where V_i is an open neighbourhood of x_i and f maps each V_i diffeomorphically onto U .*

Proof. By the inverse function theorem we can pick disjoint neighbourhoods W_i of x_i such that f maps W_i diffeomorphically onto a neighbourhood of y . Observe that $f(X - \cup_i W_i)$ is a compact set which does not contain y . Now take $U = \bigcap_i f(W_i) - f(X - \cup_i W_i)$. $\square$

If we let $\#f^{-1}(y)$ be the cardinality of $f^{-1}(y)$, the theorem implies that the function $y \mapsto \#f^{-1}(y)$ is locally constant as y ranges over regular values of f .

Theorem 1.8 (Sard's Theorem). *The set of critical values of a smooth map $f : X \rightarrow Y$ has measure zero.*

1.3 Transversality

Definition 1.9 (Transversal). A smooth map $f : X \rightarrow Y$ is said to be **transversal** to a submanifold $Z \subset Y$ if for every $x \in f^{-1}(Z)$ we have

$$\text{Image}(df_x) + T_{f(x)}Z = T_{f(x)}Y.$$

We write $f \pitchfork Z$.

Theorem 1.10 (Transversal Preimage Theorem). *If the smooth map $f : X \rightarrow Y$ is transversal to a submanifold $Z \subset Y$, then $f^{-1}(Z)$ is submanifold of X . Moreover, $f^{-1}(Z)$ and Z have the same codimension.*

An important special case occurs when f is the inclusion of a submanifold X of Y and Z is another submanifold of Y . In this case the condition of transversality reduces to

$$T_x X + T_x Z = T_x Y$$

for every $x \in X \cap Z$. If this is the case, then $X \cap Z$ is a submanifold of codimension given by

$$\text{codim}(X \cap Z) = \text{codim } X + \text{codim } Z.$$

1.4 Degree Modulo 2

Lemma 1.11 (Homotopy Lemma). *Let $f, g : X \rightarrow Y$ be smooth maps which are smoothly homotopic. Suppose X is compact, has the same dimension as Y and $\partial X = \emptyset$. If y is a regular value for both f and g , then*

$$\#f^{-1}(y) = \#g^{-1}(y) \pmod{2}.$$

Lemma 1.12 (Homogeneity Lemma). *Let X be a smooth connected manifold, possibly with boundary. Let y and z be points in $\text{Int}(X)$. Then there exists a diffeomorphism $h : X \rightarrow X$ smoothly isotopic to the identity such that $h(y) = z$.*

In what follows suppose that X is compact and without boundary and Y is connected and with the same dimension as X . Let $f : X \rightarrow Y$ be a smooth map.

Theorem 1.13 (Degree Mod 2). *If y and z are regular values of f then*

$$\#f^{-1}(y) = \#f^{-1}(z) \pmod{2}.$$

*This common residue class is called the **degree mod 2** of f , $\deg_2 f$, and only depends on the homotopy class of f .*

Corollary 1.14 (Smooth Brouwer Fixed Point Theorem). *Any smooth map $f : B^k \rightarrow B^k$ has a fixed point.*

Proof. Suppose f has no fixed point. Define $g : B^k \rightarrow S^{k-1}$ so that $g(x)$ is the point where the line segment starting at $f(x)$ passing through x hits the boundary. This is obviously smooth, and restricts to the identity on S^{k-1} .

Now the identity map on a compact boundaryless manifold has $\deg_2 = 1$, and the constant map has $\deg_2 = 0$. So they are never homotopic. This implies that there is no smooth map $f : B^k \rightarrow S^{k-1}$ which restricts to the identity on S^{k-1} , as otherwise we could construct a homotopy $H : S^k \times [0, 1] \rightarrow S^k$ between the constant map and the identity given by $H(x, t) = f(tx)$. So f must have a fixed point. $\square$

2 Length, Area and Curvature

2.1 Curves

Definition 2.1 (Curve). Let $I \subset \mathbb{R}$ be an interval and let X be a manifold. A **curve** in X is a smooth map $\alpha : I \rightarrow X$. The curve is said to be **regular** if α is an immersion, i.e., if the velocity vector $\dot{\alpha}(t) = d\alpha_t(1) \in T_{\alpha(t)}X$ is never zero.

By definition, given $t \in I$, the arc-length of $\alpha : I \rightarrow \mathbb{R}^3$ from the point t_0 is given by

$$s(t) := \int_{t_0}^t |\dot{\alpha}(\tau)| d\tau.$$

If the interval I has endpoints a and $b, a < b$, the length of α is

$$\ell(\alpha) := \int_a^b |\dot{\alpha}(t)| dt.$$

The curve is said to be parametrized by arc-length if $|\dot{\alpha}(t)| = 1$ for all $t \in I$. From now on we will assume that curves are parametrized by arc-length.

Definition 2.2. The **tangent** at $s \in I$ is $t(s) = \dot{\alpha}(s)$. The **curvature** of α at s is the number $k(s) = |\ddot{\alpha}(s)|$. The **normal vector** at s is $n(s)$, where $\ddot{\alpha}(s) = k(s)n(s)$. The **binormal vector** at s is $b(s) = t(s) \wedge n(s)$. We have $\dot{b}(s) = \tau(s)n(s)$, where $\tau(s)$ is the **torsion** at s .

Proposition 2.3 (Frenet Formulas). *We have*

$$\dot{t} = kn, \quad \dot{n} = -kt - \tau b, \quad \text{and} \quad \dot{b} = \tau n$$

Theorem 2.4 (Fundamental Theorem of the Local Theory of Curves). *Given smooth functions $k(s) > 0$ and $\tau(s), s \in I$, there exists a regular curve $\alpha : I \rightarrow \mathbb{R}^3$ such that s is arc-length, $k(s)$ is the curvature, and $\tau(s)$ is the torsion of α . Moreover any other curve $\tilde{\alpha}$, satisfying the same conditions, differs from α by an isometry.*

2.2 Isoperimetric Inequality

Lemma 2.5 (Wirtinger's Inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function which is periodic with period L . Suppose $\int_0^L f(t) dt = 0$. Then*

$$\int_0^L |f'(t)|^2 dt \geq \frac{4\pi^2}{L^2} \int_0^L |f(t)|^2 dt,$$

with equality if and only if there exist constants $a_{\pm 1}$ such that $f(t) = a_{-1}e^{-2\pi it/L} + a_1e^{2\pi it/L}$.

Theorem 2.6 (Isoperimetric Inequality in the Plane). *Let Ω be a domain, that is, a connected open set. We assume that Ω has compact closure and that its*

boundary $\partial\Omega$ is a connected 1-manifold of class C^1 . Let $A(\Omega)$ be the area of Ω . Then

$$\ell^2(\partial\Omega) \geq 4\pi A(\Omega)$$

with equality if and only if Ω is a disk.

Proof. Define the vector field $X(x, y) = (x, y)$, and let n be the outward pointing normal vector field along $\partial\Omega$. The divergence theorem gives us that

$$\int_{\Omega} \operatorname{div} X \, dA = \int_{\partial\Omega} \langle X, n \rangle \, ds.$$

But $\operatorname{div}(X) = 2$, so by Cauchy-Schwarz we have

$$2A(\Omega) \leq \int_{\partial\Omega} |X| \, ds.$$

By Cauchy-Schwarz again we have

$$\begin{aligned} 2A(\Omega) &\leq \left(\int_{\partial\Omega} |X|^2 \, ds \right)^{1/2} \left(\int_{\partial\Omega} ds \right)^{1/2} \\ &= \ell(\partial\Omega)^{1/2} \left(\int_{\partial\Omega} |X|^2 \, ds \right)^{1/2}. \end{aligned}$$

Since we parameterise $\partial\Omega$ by arc length, $X(s) = (x(s), y(s))$ along $\partial\Omega$ are C^1 and periodic with period $L = \ell(\partial\Omega)$. Hence by Wirtinger's inequality we have

$$\begin{aligned} \left(\int_{\partial\Omega} |X|^2 \, ds \right)^{1/2} &\leq \left(\frac{\ell(\partial\Omega)^2}{4\pi^2} \int_{\partial\Omega} |X'|^2 \, ds \right)^{1/2} \\ &= \frac{\ell(\partial\Omega)^{3/2}}{2\pi} \end{aligned}$$

which gives the desired result. Equality occurs if and only if we have equality in all of the above, in particular in the second $s \mapsto |X(s)|$ is constant, so Ω is a disk. $\square$

2.3 First Fundamental Form

Definition 2.7 (First Fundamental Form). Let $S \subset \mathbb{R}^3$ be a surface. The quadratic form I_p on $T_p S$ given by

$$I_p(w) := \langle w, w \rangle = |w|^2$$

is called the **first fundamental form** of the surface at p .

Definition 2.8. Two surfaces S_1 and S_2 are said to be **isometric** if there exists a diffeomorphism $f : S_1 \rightarrow S_2$ such that for all $p \in S_1$, df_p is a linear isometry between $T_p S_1$ and $T_{f(p)} S_2$.

Let $\phi : U \subset \mathbb{R}^2 \rightarrow S \subset \mathbb{R}^3$ be a parametrization of a neighbourhood of a point $p \in S$. We will denote by (u, v) points in U and let

$$\begin{aligned}\phi_u(u, v) &= \frac{\partial \phi}{\partial u}(u, v) \in T_{\phi(u, v)}S, \\ \phi_v(u, v) &= \frac{\partial \phi}{\partial v}(u, v) \in T_{\phi(u, v)}S.\end{aligned}$$

Note these are linearly independent. Set

$$\begin{aligned}E &= \langle \phi_u, \phi_u \rangle_{\phi(u, v)}, \\ F &= \langle \phi_u, \phi_v \rangle_{\phi(u, v)}, \\ G &= \langle \phi_v, \phi_v \rangle_{\phi(u, v)}.\end{aligned}$$

Since a tangent vector $w \in T_p S$ is the tangent vector of a curve $\alpha(t) = \phi(u(t), v(t))$, $t \in (-\varepsilon, \varepsilon)$, with $p = \alpha(0) = \phi(u_0, v_0)$ we have

$$\begin{aligned}I_p(\dot{\alpha}(0)) &= \langle \dot{\alpha}(0), \dot{\alpha}(0) \rangle_p \\ &= E(\dot{u})^2 + 2F\dot{u}\dot{v} + G(\dot{v})^2.\end{aligned}$$

We can compute the length of a curve in S then by integrating $\sqrt{E(\dot{u})^2 + 2F\dot{u}\dot{v} + G(\dot{v})^2}$. Note also that $|\phi_u \wedge \phi_v| = \sqrt{EG - F^2}$.

Definition 2.9 (Area). Let $\Omega \subset S$ be a bounded domain contained in the image of a parametrization $\phi : U \rightarrow S$. The positive number

$$A(\Omega) = \int_{\phi^{-1}(\Omega)} |\phi_u \wedge \phi_v| \, du \, dv$$

is called the area of Ω .

2.4 The Gauss Map

Given a parametrization $\phi : U \subset \mathbb{R}^2 \rightarrow S \subset \mathbb{R}^3$ around a point $p \in S$, we can choose a unit normal vector at each point of $\phi(U)$ by setting

$$N(q) = \frac{\phi_u \wedge \phi_v}{|\phi_u \wedge \phi_v|}(q).$$

Definition 2.10 (Orientable). A surface $S \subset \mathbb{R}^3$ is **orientable** if it admits a smooth field of unit normal vectors. The choice of such a field is called an **orientation**.

Definition 2.11 (Gauss Map). Let S be an oriented surface and $N : S \rightarrow S^2$ the smooth field of unit normal vectors defining the orientation. The map N is called the **Gauss map** of S .

Since $T_p S$ and $T_{N(p)} S^2$ are parallel planes, we will regard dN_p as a linear map $dN_p : T_p S \rightarrow T_p S$.

Proposition 2.12. *The linear map $dN_p : T_p S \rightarrow T_p S$ is self-adjoint.*

Proof. Let $\phi : U \rightarrow S$ be a parametrization around p . If $\alpha(t) = \phi(u(t), v(t))$ is a curve in $\phi(U)$ with $\alpha(0) = p$ we have

$$\begin{aligned} dN_p(\dot{\alpha}(0)) &= dN_p(\dot{u}(0)\phi_u + \dot{v}(0)\phi_v) \\ &= \left. \frac{d}{dt} \right|_{t=0} N(u(t), v(t)) \\ &= \dot{u}(0)N_u + \dot{v}(0)N_v \end{aligned}$$

In particular $dN_p(\phi_u) = N_u$ and $dN_p(\phi_v) = N_v$ and since $\{\phi_u, \phi_v\}$ is a basis of the tangent plane, we only have to prove that

$$\langle N_u, \phi_v \rangle = \langle N_v, \phi_u \rangle$$

To prove the last equality, observe that $\langle N, \phi_u \rangle = \langle N, \phi_v \rangle = 0$. Take derivatives with respect to v and u to obtain:

$$\begin{aligned} \langle N_v, \phi_u \rangle + \langle N, \phi_{uv} \rangle &= 0, \\ \langle N_u, \phi_v \rangle + \langle N, \phi_{vu} \rangle &= 0 \end{aligned}$$

which gives the desired equality. $\square$

Definition 2.13 (Second Fundamental Form). The quadratic form defined on $T_p S$ by $II_p(w) = -\langle dN_p(w), w \rangle$ is called the **second fundamental form** of S at p .

Definition 2.14 (Normal Curvature). Let $\alpha : (-\varepsilon, \varepsilon) \rightarrow S$ be a curve, $\alpha(0) = p$. Then the **normal curvature** of α at p is defined by $k_n(p) = \langle N, kn \rangle$ where N is the Gauss map, k is the curvature of α and n is the unit normal to α at p (i.e. $kn = \ddot{\alpha}$).

Proposition 2.15. $k_n(p) = II_p(\dot{\alpha}(0))$.

Definition 2.16 (Principal Curvatures and Directions). As $dN_p : T_p S \rightarrow T_p S$ is self adjoint, it can be diagonalised. Let $e_1, e_2 \in T_p S$ be such that, with respect to this basis, we have

$$dN_p = \begin{pmatrix} -k_1 & 0 \\ 0 & -k_2 \end{pmatrix}$$

where $k_1 \geq k_2$. We call k_1, k_2 the **princial curvatures**, and e_1, e_2 the **principal directions**.

From standard linear algebra we get that k_1 (respectively k_2) is the maximum (minimum) value of II_p on the set of unit vectors in $T_p S$. That is, they are the extreme values of the normal curvature at p .

Definition 2.17 (Gaussian and Mean Curvature). The determinant of dN_p is the **Gaussian curvature** $K(p)$ of S at p . Minus half of the trace of dN_p is the **mean curvature** $H(p)$ of S at p .

Clearly $K = k_1 k_2$ and $H = \frac{k_1 + k_2}{2}$.

A point $p \in S$ of a surface is called **elliptic** if $K(p) > 0$, **hyperbolic** if $K(p) < 0$, **parabolic** if $K(p) = 0$ and $dN_p \neq 0$, and **planar** if $dN_p = 0$. A point $p \in S$ is called **umbilical** if $k_1 = k_2$.

2.5 Local Coordinates

Let $\phi : U \rightarrow S$ be a parametrization around a point $p \in S$. Let us express the second fundamental form in the basis $\{\phi_u, \phi_v\}$. Since $\langle N, \phi_u \rangle = \langle N, \phi_v \rangle = 0$ we have

$$\begin{aligned} e &= -\langle N_u, \phi_u \rangle = \langle N, \phi_{uu} \rangle, \\ f &= -\langle N_v, \phi_u \rangle = \langle N, \phi_{uv} \rangle = -\langle N_u, \phi_v \rangle, \\ g &= -\langle N_v, \phi_v \rangle = \langle N, \phi_{vv} \rangle. \end{aligned}$$

If α is a curve passing at $t = 0$ through p we can write:

$$\begin{aligned} II_p(\dot{\alpha}(0)) &= -\langle dN_p(\dot{\alpha}(0)), \dot{\alpha}(0) \rangle \\ &= e(\dot{u})^2 + 2f\dot{u}\dot{v} + g(\dot{v})^2. \end{aligned}$$

With respect to the basis ϕ_u, ϕ_v , we can express dN_p as a matrix, namely

$$\begin{aligned} dN_p(\phi_u) &= N_u = a_{11}\phi_u + a_{21}\phi_v \\ dN_p(\phi_v) &= N_v = a_{12}\phi_u + a_{22}\phi_v \end{aligned}$$

Taking inner products of the above equations with ϕ_u, ϕ_v we get

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = - \begin{pmatrix} e & f \\ f & g \end{pmatrix}$$

But with respect to the basis ϕ_u, ϕ_v , dN_p has matrix $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$. Linear algebra then gives

Corollary 2.18. *We can write*

$$K = \frac{eg - f^2}{EG - F^2}, \text{ and } H = \frac{eG - 2fF + gE}{2(EG - F^2)}.$$

2.6 Theorema Egregium

Theorem 2.19 (Theorema Egregium). *The Gaussian curvature K of a surface is invariant under isometries.*

Proof. It suffices to write K in terms only of the coefficients E, F, G of the first fundamental form and their derivatives. Let $\phi : U \rightarrow S$ be a parameterisation. Then at each point in the image we have a basis of $\mathbb{R}^3$ given by $\{\phi_u, \phi_v, N\}$. We can then express the derivatives of ϕ_u and ϕ_v in this basis:

$$\begin{aligned}\phi_{uu} &= \Gamma_{11}^1 \phi_u + \Gamma_{11}^2 \phi_v + eN, \\ \phi_{uv} &= \Gamma_{12}^1 \phi_u + \Gamma_{12}^2 \phi_v + fN, \\ \phi_{vu} &= \Gamma_{21}^1 \phi_u + \Gamma_{21}^2 \phi_v + fN, \\ \phi_{vv} &= \Gamma_{22}^1 \phi_u + \Gamma_{22}^2 \phi_v + gN,\end{aligned}$$

where the Γ_{ij}^k are the **Christoffel symbols**. Taking inner products with ϕ_u and ϕ_v , we can see that we can solve for the Christoffel symbols in terms of E, F, G and their derivatives. So Christoffel symbols are invariant under isometries.

Consider $\phi_{uuv} = \phi_{uvu}$, and differentiating our previous expressions and substituting in gives (after some manipulation)

$$\begin{aligned} & (\Gamma_{12}^2)_u - (\Gamma_{11}^2)_v + \Gamma_{12}^1 \Gamma_{11}^2 + \\ & \quad \Gamma_{12}^2 \Gamma_{12}^2 - \Gamma_{11}^2 \Gamma_{22}^2 - \Gamma_{11}^1 \Gamma_{12}^2 \\ &= -fa_{21} + ea_{22} = -E \frac{eg - f^2}{EG - F^2} = -EK. \end{aligned}$$

Thus K can be expressed solely in terms of the coefficients of the first fundamental form and their derivatives as required. $\square$

Definition 2.20 (Isothermal Parameterisation). A parameterization is **isothermal** if $E = G = \lambda^2(u, v)$ and $F = 0$.

Proposition 2.21. For isothermal parameterization, $K = -\frac{1}{\lambda^2} \Delta(\log \lambda)$, where Δ is the Laplacian in (u, v) -coordinates.

3 Geodesics & Minimal Surfaces

3.1 Geodesics

Let $S \subseteq \mathbb{R}^3$ be a surface with $p, q \in S$. Let $\Omega(p, q)$ be the set of all curves $\alpha : [0, 1] \rightarrow S$ with $\alpha(0) = p$ and $\alpha(1) = q$.

Definition 3.1 (Energy Functional). The **energy** $E : \Omega(p, q) \rightarrow \mathbb{R}$ is given by

$$E(\alpha) = \frac{1}{2} \int_0^1 |\dot{\alpha}|^2 dt.$$

Let $\alpha_s \in \Omega(p, q)$ be a smooth one parameter family of curves, with $s \in (-\varepsilon, \varepsilon)$. Let $E(s) = E(\alpha_s)$. Then we have that

$$\frac{dE}{ds} = \int_0^1 \left\langle \frac{\partial}{\partial s} \frac{\partial \alpha_s}{\partial t}, \frac{\partial \alpha_s}{\partial t} \right\rangle dt$$

Integrating by parts we get

$$\begin{aligned} \left. \frac{dE}{ds} \right|_{s=0} &= \langle J(1), \dot{\alpha}(1) \rangle - \langle J(0), \dot{\alpha}(0) \rangle \\ &\quad - \int_0^1 \langle J(t), \ddot{\alpha}(t) \rangle dt \end{aligned}$$

where

$$J(t) = \left. \frac{\partial \alpha_s(t)}{\partial s} \right|_{s=0}$$

Since $\alpha_s \in \Omega(p, q)$, $J(0) = J(1) = 0$. So we get that

$$\left. \frac{dE}{ds} \right|_{s=0} = - \int_0^1 \langle J(t), \ddot{\alpha}(t) \rangle dt$$

Now notice that for each $t \in [0, 1]$, $J(t) \in T_{\alpha(t)}S$, since $s \mapsto \alpha_s(t)$ is a curve in S . So if α is such that $\ddot{\alpha} \perp T_{\alpha(t)}S$ for all t , then α extremises E .

Definition 3.2 (Geodesic). A curve $\alpha : I \rightarrow S$ is a **geodesic** if for all $t \in I$, $\ddot{\alpha}(t)$ is orthogonal to $T_{\alpha(t)}S$.

3.2 Covariant Derivative

Definition 3.3 (Vector Field). Let $\alpha : I \rightarrow S$ be a curve. A **vector field** along α is a smooth map $V : I \rightarrow \mathbb{R}^3$ such that for all t , $V(t) \in T_{\alpha(t)}S$.

Definition 3.4 (Covariant Derivative). The **covariant derivative** of a vector field V along α is

$$\frac{DV}{dt}(t) = \text{proj}_{T_{\alpha(t)}S} \left(\frac{dV}{dt} \right)$$

where $\text{proj}_{T_{\alpha(t)}S}$ is the orthogonal projection onto $T_{\alpha(t)}S$.

Proposition 3.5. A curve α is a geodesic if and only if $\frac{D\dot{\alpha}}{dt} = 0$ for all t .

Definition 3.6 (Parallel). A vector field V along α is **parallel** if $\frac{DV}{dt} = 0$.

Proposition 3.7. Let V, W be parallel vector fields along α . Then $\langle V(t), W(t) \rangle$ is constant.

Corollary 3.8. If α is a geodesic, then $|\dot{\alpha}|$ is constant. So geodesics are parametrised proportional to arc length.

3.3 Local Coordinates

Let $\phi : U \rightarrow S$ be a parametrisation, $\alpha : I \rightarrow S$ a curve, with $\alpha(I) \subseteq \phi(U)$. Write $\alpha(t) = \phi(u(t), v(t))$. Let V be a vector field along α . Then there are functions $a(t), b(t)$ such that

$$V(t) = a(t)\phi_u + b(t)\phi_v$$

Differentiating this, we get that

$$\frac{dV}{dt} = a(\phi_{uv}\dot{u} + \phi_{uv}\dot{v}) + b(\phi_{vu}\dot{u} + \phi_{vv}\dot{v}) + a\phi_u + b\phi_v$$

The covariant derivative is just the ϕ_j and ϕ_v components of this, since N is orthogonal to $T_{a(t)}S$. Therefore, in terms of Christoffel symbols, we have that

$$\begin{aligned} \frac{DV}{dt} &= (\dot{a} + a\dot{u}\Gamma_{11}^1 + a\dot{v}\Gamma_{12}^1 + b\dot{u}\Gamma_{12}^1 + b\dot{v}\Gamma_{22}^1)\phi_u \\ &\quad + (b + a\dot{u}\Gamma_{11}^2 + a\dot{v}\Gamma_{12}^2 + b\dot{u}\Gamma_{12}^2 + b\dot{v}\Gamma_{22}^2)\phi_v \end{aligned}$$

From this expression, we see that the covariant derivative only depends on the first fundamental form.

Proposition 3.9 (Geodesic Equations). $\alpha(t) = \phi(u(t), v(t))$ is a geodesic if and only if

$$\begin{aligned} \ddot{u} + \Gamma_{11}^1\dot{u}^2 + 2\Gamma_{12}^1\dot{u}\dot{v} + \Gamma_{22}^1\dot{v}^2 &= 0 \\ \ddot{v} + \Gamma_{11}^2\dot{u}^2 + 2\Gamma_{12}^2\dot{u}\dot{v} + \Gamma_{22}^2\dot{v}^2 &= 0 \end{aligned}$$

Proposition 3.10 (Parallel Transport). Given $v_0 \in T_{\alpha(t_0)}S$, there exists a unique parallel vector field $V(t)$ along $\alpha(t)$, with $V(t_0) = v_0$. We call $V(t_1)$ the **parallel transport** of v_0 along α at t_1 .

Corollary 3.11 (Geodesic Existence). Given $p \in S, v \in T_pS$, there exists $\varepsilon > 0$, and a unique geodesic $\gamma : (-\varepsilon, \varepsilon) \rightarrow S$ such that $\gamma(0) = p$ and $\dot{\gamma}(0) = v$.

Definition 3.12. Let $\alpha \in \Omega(p, q)$. Define $P : T_pS \rightarrow T_qS$ the map sending $v \in T_pS$ to the **parallel transport** of v along α at q .

3.4 Exponential Map

Proposition 3.13. Given $p \in S, v \in T_pS$, let $\gamma_v : (-\varepsilon, \varepsilon) \rightarrow S$ by the unique geodesic with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. Then $\gamma_{\lambda v}$ is defined on $(-\varepsilon/\lambda, \varepsilon/\lambda)$. Furthermore, $\gamma_{\lambda v}(t) = \gamma_v(\lambda t)$.

Definition 3.14 (Exponential Map). Suppose $v \in T_pS$ nonzero is such that $\gamma_v(1)$ is defined, we define the **exponential map** $\exp_p(v) = \gamma_v(1)$.

We note exists $\varepsilon > 0$ such that $\exp_p : B_\varepsilon(0) \rightarrow S$ is well defined and smooth.

Proposition 3.15. If S is closed, then $\exp_p$ is defined on all of T_pS .

Proposition 3.16. $\exp_p : B_\varepsilon(0) \rightarrow S$ is a diffeomorphism onto its image in a neighbourhood $U \subseteq B_\varepsilon(0)$ of $0 \in T_pS$

Proof. By the inverse function theorem, suffices to show $d(\exp_p)_0$ is nonsingular. Let $\alpha(t) = tv$ for some fixed $v \in T_pS$. Then $\exp_p(tv) = \gamma_v(t)$ at $t = 0$ has tangent vector v . So $d(\exp_p)_0(v) = v$. $\square$

Definition 3.17 (Normal Neighbourhood). Let U be as in the previous proposition. Then $V = \exp_p(U)$ is called a **normal neighbourhood** of p .

Corollary 3.18. $\exp_p : U \rightarrow V$ is a parametrisation.

Proposition 3.19. If we choose cartesian coordinates on $T_p S$, then with the $\exp_p$ parametrisation, we have the first fundamental form

$$E(p) = G(p) = 1 \text{ and } F(p) = 0$$

Definition 3.20 (Geodesic Polars). If we choose polar coordinates (r, θ) for $T_p S$, then we have the **geodesic polar coordinates**. That is,

$$\begin{aligned} \phi(r, \theta) &= \exp_p(r(\cos(\theta)e_1 + \sin(\theta)e_2)) \\ &= \exp_p(rv(\theta)) = \gamma_{v(\theta)}(t) \end{aligned}$$

where $v(\theta) = \cos(\theta)e_1 + \sin(\theta)e_2$.

Definition 3.21 (Geodesic Circles, Radial Geodesics). The images of circles centred in the origin under the map ϕ are called **geodesic circles** (i.e. $r = \text{const}$). Similarly, the images of lines through the origin (i.e. $\theta = \text{const}$) are called **radial geodesics**.

Proposition 3.22. For geodesic polars we have

$$\begin{aligned} E &= 1, \quad F = 0, \quad G(0, \theta) = 0 \\ \text{and} \quad (\sqrt{G})_r(0, \theta) &= 1 \end{aligned}$$

Moreover, the Gaussian curvature can be written as

$$K = -\frac{(\sqrt{G})_{rr}}{\sqrt{G}}$$

Proof. By definition of ϕ , we have that $\phi_r = \dot{\gamma}_{v(\theta)}(r)$, so $E = 1$ as $v(\theta)$ is a unit vector and geodesics travel at constant speed. Now let $w = \frac{dv}{d\theta}$. Then by chain rule, we have that

$$\phi_\theta = d(\exp_p)_{rv}(rw) = r d(\exp_p)_{rv}(w)$$

So we have that

$$\begin{aligned} F &= r \left\langle \dot{\gamma}_v(r), d(\exp_p)_{rv}(w) \right\rangle \\ G &= r^2 \left| d(\exp_p)_{rv}(w) \right|^2 \end{aligned}$$

Clearly $F(0, \theta) = 0$, and from the last equality, we find that

$$(\sqrt{G})_r(0, \theta) = \left| d(\exp_p)_0(w) \right| = |w| = 1$$

Finally, we can compute

$$\begin{aligned}
F_r &= \langle \phi_{rr}, \phi_\theta \rangle + \langle \phi_r, \phi_{\theta r} \rangle \\
&= \langle \phi_r, \phi_{\theta r} \rangle \\
&= \frac{1}{2} \frac{\partial}{\partial \theta} \langle \phi_r, \phi_r \rangle \\
&= \frac{1}{2} E_\theta \\
&= 0
\end{aligned}$$

where we used the fact that $\phi(\cdot, \theta) = \gamma_v$ is a geodesic, so $\phi_{rr} = \ddot{\gamma}_v$ is normal to $T_p S$. So $F = 0$ identically. We omit the computation for K , and note that it can be computed using Christoffel symbols. $\square$

3.5 Geodesic Curvature

Definition 3.23 (Algebraic Value of the Covariant Derivative). Let W be a differentiable field of unit vectors along a curve $\alpha : I \rightarrow S$ along an oriented surface S . Then

$$\left[\frac{DW}{dt} \right] = \left\langle \frac{dW}{dt}, N \wedge W \right\rangle$$

Proposition 3.24. Let W be a field of unit vectors along α . Then $\frac{DW}{dt}$ is parallel to $N \wedge W$, and we have that

$$\frac{DW}{dt} = \left[\frac{DW}{dt} \right] (N \wedge W)$$

Definition 3.25 (Geodesic Curvature). Let $\alpha : I \rightarrow S$ be a regular curve parametrised by arc length. The algebraic value of the covariant derivative

$$\kappa_g(s) = \left[\frac{D\dot{\alpha}}{dt} \right] = \langle \ddot{\alpha}, N \wedge \dot{\alpha} \rangle$$

is called the **geodesic curvature** of α at $\alpha(s)$.

Proposition 3.26. α is a geodesic if and only if its geodesic curvature is identically zero.

Proposition 3.27. Let k and n be the curvature and unit normal for α . Then we have that

$$\ddot{\alpha} = k_n N + k_g (N \wedge \dot{\alpha})$$

where κ_n, κ_g are the normal and geodesic curvatures respectively.

Proof. Since W has norm 1, we have that $\langle W, W \rangle = 1$, so $\langle \frac{dW}{dt}, W \rangle = 0$. Hence $\frac{dW}{dt}$ is perpendicular to W . Thus, $\frac{DW}{dt}$ must be perpendicular to both W and N , so it is parallel to $N \wedge W$. $\square$

Definition 3.28 (Perpendicular Vector Field). Let V be a unit vector field along $\alpha : I \rightarrow S$. Let $iV(t)$ be the unique vector field along α such that for every $t \in I$, $V(t), iV(t), N(t)$ forms a positively oriented orthonormal basis of $\mathbb{R}^3$. That is,

$$V(t) \wedge iV(t) = N(t)$$

Proposition 3.29. Let V, W be unit vector fields along $\alpha : I \rightarrow S$. Then there exists smooth functions a, b , such that

$$W(t) = a(t)V(t) + b(t)iV(t)$$

with $a^2 + b^2 = 1$. Furthermore, if we fix $t_0 \in I$, and let φ_0 be such that

$$a(t_0) = \cos(\varphi_0) \quad \text{and} \quad b(t_0) = \sin(\varphi_0)$$

then there exists a smooth function $\varphi : I \rightarrow S$ such that

$$a(t) = \cos(\varphi(t)), \quad b(t) = \sin(\varphi(t)) \quad \text{and} \quad \varphi(t_0) = \varphi_0$$

Proof. $V(t), iV(t)$ is an orthonormal basis of $T_{\alpha(t)}S$. The construction of φ is as in the construction of the winding number in Complex Analysis. $\square$

Definition 3.30 (Smooth Determination of Angle). φ from the previous proposition is called a **smooth determination of the angle** from V to W .

Proposition 3.31. Let V, W be unit vector fields along $\alpha : I \rightarrow S$ and φ by a smooth determination of angle from V to W . Then

$$\left[\frac{DW}{dt} \right] - \left[\frac{DV}{dt} \right] = \frac{d\varphi}{dt}$$

Proposition 3.32. Let $\alpha : I \rightarrow S$ be a curve parametrised by arc length, $V(s)$ a parallel unit vector field along α , φ a smooth determination of angle from V to $\dot{\alpha}$. Then

$$\kappa_g(s) = \frac{d\varphi}{ds}$$

Proof. $\left[\frac{DV}{dt} \right] = 0$ as V is parallel. $\square$

4 Gauss-Bonnet

Theorem 4.1 (Gauss's Theorem for Geodesic Triangles). Let T be a geodesic triangle on a surface S . Suppose T is small enough so that it is contained in a normal neighbourhood of one of its vertices, then

$$\int_T K \, dA = \alpha_1 + \alpha_2 + \alpha_3 - \pi$$

where K is the Gaussian curvature of S , and $0 < \alpha_i < \pi$ are the internal angles of T .

Proof. We can assume without loss of generality that we have geodesic polar coordinates centred at one of the vertices of T , one of the edges corresponds to $\theta = 0$ and another corresponds to $\theta = \theta_0$. The remaining edge is a geodesic segment γ .

First notice that γ can be written in the form $r = h(\theta)$. Suppose not, then there exists such that $\dot{\gamma}(s)$ is parallel to ϕ_r . But radial segments are geodesics, so this means that γ is radial. Contradiction. Hence we can write γ as $r = h(\theta)$. Then

$$\begin{aligned} \int_T K \, dA &= \int_T K \sqrt{G} \, dr \, d\theta \\ &= \int_0^{\theta_0} \left(\lim_{\varepsilon \rightarrow 0} \int_\varepsilon^{h(\theta)} K \sqrt{G} \, dr \right) d\theta \end{aligned}$$

But in geodesic polar coordinates, we have $K \sqrt{G} = -(\sqrt{G})_{rr}$, and $\lim_{r \rightarrow 0} (\sqrt{G})_r = 1$, so

$$\lim_{\varepsilon \rightarrow 0} \int_\varepsilon^{h(\theta)} K \sqrt{G} \, dr = 1 - (\sqrt{G})_r(h(\theta), \theta)$$

Now suppose $\gamma(s) = \phi(r(s), \theta(s))$ makes an angle $\varphi(s)$ with ϕ_r , that is, the curves $\theta = \text{const}$. Then the previous corollary ($u = r, v = \theta$) gives that

$$(\sqrt{G})_r \frac{d\theta}{ds} + \frac{d\varphi}{ds} = 0$$

as γ is a geodesic. Therefore, we have that

$$\begin{aligned} \int_T K \, dA &= \int_0^{\theta_0} \left(1 - (\sqrt{G})_r(h(\theta), \theta) \right) d\theta \\ &= \int_0^{\theta_0} d\theta - \int_0^{s_0} (\sqrt{G})_r \frac{d\theta}{ds} ds \\ &= \theta_0 + \int_0^{s_0} \frac{d\varphi}{ds} ds \\ &= \theta_0 + \int_{\varphi(0)}^{\varphi(s_0)} d\varphi \\ &= \theta_0 + \varphi(s_0) - \varphi(0) \end{aligned}$$

Finally, by the orientations, we have the result. $\square$

Definition 4.2 (Triangulation). Let S be a compact surface. A **triangulation** of S is a finite number of closed subsets $T_1, \dots, T_n$ which cover S , each T_i is homeomorphic to a Euclidean triangle in the plane. Moreover, any two distinct triangles are either disjoint, share a vertex, or share an edge.

Theorem 4.3. *Triangulations always exist. Furthermore, we can choose it so that each T_i is diffeomorphic to a Euclidean triangle, and each edge is a geodesic segment.*

Definition 4.4 (Euler Characteristic). Given a triangulation of S , let F be the number of faces, E the number of edges, V the number of vertices. Then

$$x(S) = V - E + F$$

is the **Euler characteristic** of S .

This is independent of the choice of triangulation.

Proposition 4.5 (Classification of Compact Orientable Surfaces). *All compact orientable surfaces are diffeomorphic to some Σ_g where g is a g -holed torus. g is called the **genus** of Σ_g . Furthermore,*

$$\chi(\Sigma_g) = 2 - 2g$$

Theorem 4.6 (Global Gauss-Bonnet). *Let S be a compact surface without boundary. Then*

$$\int_S K dA = 2\pi\chi(S)$$

Proof. Consider a triangulation by geodesic triangles $T_1, \dots, T_F$. We can assume wlog that each T_i is contained in a normal neighbourhood of one of its vertices. Let $\alpha_i, \beta_i, \gamma_i$ be the interior angles of T_i . Then by Gauss's theorem for triangles, we have that

$$\int_{T_i} K dA = \alpha_i + \beta_i + \gamma_i - \pi$$

Summing over all i , we have that

$$\int_S K dA = \sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) - \pi F$$

Now notice that the sum of the angles at every vertex is 2π , so

$$\sum_{i=1}^F (\alpha_i + \beta_i + \gamma_i) = 2\pi V$$

Finally, for a triangulation, every edge belongs to two triangles, so $2E = 3F$. Putting this all together we get that

$$\int_S K dA = \pi(2V - F) = 2\pi\chi(S).$$

□

Theorem 4.7 (Local Gauss-Bonnet). *Let $\phi : U \rightarrow S$ be an orthogonal parametrisation of an oriented surface S , U is a disc in $\mathbb{R}^2$, and ϕ is compatible with the orientation of S . Let $\alpha : I \rightarrow \phi(U)$ be a smooth simple closed curve enclosing*

a domain R . Suppose α is positively oriented and parametrised by arc length. Then

$$\int_l k_g(s)ds + \int_R K dA = 2\pi$$

where k_g is the geodesic curvature of α .

Theorem 4.8 (Gauss-Bonnet with Boundary). *Let $R \subseteq S$ be a connected open relatively compact² subset. Suppose ∂R contains of n piecewise smooth simple closed curves $\alpha_i : I_i \rightarrow S$, where the images do not intersect. Suppose the α_i are parametrised by arc length, and are positively oriented. Let θ_i be the external angles of the vertices of these curves. Then*

$$\sum_{i=1}^n \int_{l_i} k_g(s)ds + \int_R K dA + \sum_i \theta_i = 2\pi\chi(R)$$

Theorem 4.9. *Suppose S is a compact orientable surface with $K > 0$. Then S is diffeomorphic to S^2 . Moreover, if α, β are simple closed geodesics on S , then they must intersect.*

Proof. Gauss-Bonnet gives us that $\chi(S) > 0$, so S is diffeomorphic to S^2 . Now suppose α, β do not intersect. Then they bound a domain R with $\chi(R) = 0$. But then Gauss-Bonnet means that R must in fact have measure zero. Contradiction. $\square$

4.1 Minimal Surfaces

Definition 4.10 (Minimal Surface). A surface S is **minimal** if its mean curvature vanishes everywhere.

Definition 4.11 (Normal Variation). Let $\phi : U \rightarrow S$ be a parametrisation, $D \subseteq U$ bounded open connected, with $\bar{D} \subseteq U$. Let $h : \bar{D} \rightarrow \mathbb{R}$ be smooth. Then the **normal variation** of $\phi(\bar{D})$ determined by h is the map $\rho : \bar{D} \times (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^3$ given by

$$\rho(u, v, t) = \phi(u, v) + th(u, v)N(u, v)$$

²That is, the closure is compact.